

Shubhranshu

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PROFILE

Fresher CSE graduate who enjoys building things on both sides of the stack – from training deep learning models on clinical datasets to shipping live React and Next.js frontends. Gained hands-on experience at ISRO and MNNIT Allahabad, built and deployed full-stack AI apps with Flask and FastAPI backends on my own, and shipped interactive BI dashboards with Power BI and Tableau. Comfortable switching between Python, JavaScript, and SQL depending on what the problem calls for.

TECHNICAL SKILLS

ML & CV	PyTorch, TensorFlow, Keras, Scikit-learn, PyTorch Geometric, OpenCV
Frontend	React.js, Next.js, Tailwind CSS, SCSS, Three.js, Framer Motion
Backend	Flask, FastAPI, Node.js, Express.js, REST APIs
Data & BI	Power BI, Tableau, Pandas, NumPy, Matplotlib, Seaborn, NLTK, VADER
Languages	Python, JavaScript, SQL, HTML5, CSS3
Databases	MongoDB, MySQL, Snowflake
Tools & Deploy	Docker, Git, Vercel, Render, HuggingFace Spaces, Azure ML Studio
Concepts	CNNs, U-Net, GNNs, NLP, Data Modeling, DAX, REST API Design, Docker

PROJECTS

PolypSeg AI – Full-Stack Medical Image Segmentation | [Next.js](#), [Flask](#), [TensorFlow](#), [Docker](#) | [Live Site](#) | [GitHub](#)

- Built end-to-end: a Next.js 14/Tailwind/Three.js frontend with a 3D animated hero and image-upload demo panel, paired with a Flask REST API serving an Attention U-Net (76.81% Dice score, 69.01% IoU on CVC-ClinicDB).
- Containerized the backend with Docker on HuggingFace Spaces and deployed the frontend on Vercel – two independent deployments connected via a configurable API endpoint.
- Wrote setup and API documentation so the project is reproducible end-to-end, from model weights to a running live demo.

GNN for Survival Prediction | [PyTorch](#), [PyTorch Geometric](#) | [GitHub](#)

- Modeled the Titanic dataset as a k-NN graph (891 nodes, 1,200 edges) and trained a Graph Convolutional Network, reaching an 85% concordance index and 0.6942 F1-score.
- Cut prediction error by 15% over baseline models by training with GPU acceleration over 10,000 iterations.
- Compared graph-based modeling against traditional tabular baselines to evaluate whether relational structure in the data added predictive value.

Handwritten Digit Recognition – Full-Stack App | [React](#), [SCSS](#), [FastAPI](#), [TensorFlow](#) | [Live Site](#) | [GitHub](#)

- Trained a CNN on MNIST to 98.74% validation accuracy; built a 10-step OpenCV preprocessing pipeline and served the model via FastAPI on Render, consumed by a React/SCSS canvas UI deployed on Vercel.
- Styled the app with a custom parallax scroll effect and a fully responsive layout, keeping the drawing canvas usable across desktop and mobile.

Personal Portfolio Website | [Next.js](#), [Tailwind CSS](#), [Framer Motion](#) | [shubhranshu.xyz](#) | [GitHub](#)

- Designed and built a personal portfolio with Framer Motion animations, a Resend-powered contact form, and CI/CD from GitHub to Vercel; live at shubhranshu.xyz with a custom domain and private email.
- Integrated the custom domain via Namecheap and configured private email, handling both the technical and non-technical side of shipping a personal brand online.

Classic Models Sales Dashboard | [Power BI](#), [MySQL](#) | [GitHub](#)

- Built an interactive Power BI dashboard tracking revenue margins, product line KPIs, and geographic sales distribution using relational MySQL data.
- Designed a normalized MySQL schema to feed the report layer, enabling drill-down analysis from company-wide revenue down to individual product lines.

Washington EV Population Dashboard | [Tableau](#) | [Live Dashboard](#)

- Built an interactive Tableau dashboard analyzing EV adoption trends, battery range vs. MSRP, and county-level geographic distribution across Washington State.
- Used filters and parameter controls so viewers can slice the data by county, manufacturer, or vehicle type without needing to touch the underlying data.

Amazon Sentiment Analysis | [Python](#), [NLTK](#), [VADER](#) | [GitHub](#)

- Used VADER's rule-based sentiment scoring to classify 10,000 Amazon Fine Food reviews as positive or negative, reaching 88% accuracy and a 0.85 F1-score.
- Trimmed the feature space by 40% through tokenization and stop-word removal, keeping the pipeline lightweight and interpretable.

Adult Income Prediction | [Python](#), [Scikit-learn](#), [Pandas](#) | [GitHub](#)

- Built an end-to-end ML pipeline on the UCI Adult Census dataset (48K records) comparing Random Forest, Gradient Boosting, and Logistic Regression with cross-validation and ROC-AUC scoring.
- Wrapped all preprocessing (scaling, encoding, imputation) into a single reusable sklearn Pipeline using ColumnTransformer, making the model easy to retrain on new data.

INTERNSHIPS

Machine Learning InternJun 2024 – Jul 2024 (Prayagraj, UP)

Department of CSE, MNNIT Allahabad

- Built an Attention U-Net using TensorFlow and Keras for polyp segmentation on 1,500 CVC-ClinicDB colonoscopy images, achieving 95.20% precision, 81.50% accuracy, and a 0.87 F1-score.
- Built an OpenCV preprocessing pipeline (resize, normalize, augment, mask align), reducing processing time by 20% and improving diagnostic accuracy by 10% over traditional methods.

AI/ML for Geodata AnalysisAug 2024 (Remote)

ISRO | Indian Institute of Remote Sensing (IIRS), Dehradun

- Applied ML and CV techniques to satellite imagery for geodata processing and spatial feature extraction under ISRO's Department of Space, Government of India.
- Worked on predictive modeling and feature extraction from multispectral geospatial datasets using Python.
- Collaborated remotely with a government research team, following institutional data-handling and reporting standards throughout the project.

Biostatistics InternSep 2024 – Oct 2024 (Remote)

Teachnook

- Applied statistical inference, hypothesis testing, and ML in Python for survival prediction on biostatistical datasets, achieving a 0.6942 F1-score.
- Conducted data visualization and biological dataset analysis using Pandas, Matplotlib, and Seaborn.
- Interpreted model outputs in the context of biological significance, not just statistical accuracy, to make results usable for non-technical stakeholders.

Full Stack Development TraineeAug 2022 – Sep 2022 (Prayagraj, UP)

Allsoft Solutions | IBM Business Partner

- Completed an IBM Business Partner-certified, industry-oriented training program on full-stack web development, building Assignment 0 using HTML, CSS, JavaScript, Node.js, and MongoDB.
- Gained hands-on experience with server-side JavaScript (Node.js) and NoSQL data modeling (MongoDB) (Ref. MO/SEP2022/E10203).

EDUCATION

- United College of Engineering and Research

Bachelor of Technology – Computer Science & EngineeringCGPA: 6.8

Prayagraj, UP2021 – 2025

CERTIFICATIONS

Coursera	Machine Learning Pipelines with Azure ML Studio (May 2026)
Udemy	Data Analysis – SQL, Tableau, Power BI & Excel (May 2026)
Coursera/UC Davis	SQL for Data Science (Aug 2025)
Coursera	Getting Started with Microsoft Excel (Jul 2025)
Accenture/Forage	Data Analytics & Visualization Job Simulation (Jul 2024)
Google Cloud	Using Beam ML to catch Toxicity in Gaming (Nov 2023)
HackerRank	SQL (Advanced)